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15 August 1958
Superseding
MIL-R-9522(USAF)
18 May 1956

MILITARY SPECIFICATION

BEACON, RADIO AN/MRN-13( )

1. SCOPE

1.1 This specification covers one type of vehicular-mounted, air transportable low frequency beacon designated Beacon, Radio AN/MRN-13( ) hereinafter referred to as "the radio beacon."

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-E-4158	Electronic Equipment, Ground, General Requirements For
MIL-T-4807	Tests; Vibration and Shock, Ground Electronic Equipment (Requirements For)
MIL-D-5028	Drawings and Data Lists, Preparation of Manufacturer's (For Production Aeronautical and Associated Equipment)
MIL-E-5272	Environmental Testing Aeronautical and Associated Equipment, General Specification For
MIL-P-7936	Parts and Equipment, Aeronautical Preparation For Delivery
MIL-T-9107	Test Reports, Preparation Of

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DRAWINGS

Craig System

72R70 Beacon, Radio AN/MRN-13( )

(Copies of applicable documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 COMPONENTS.- The radio beacon shall consist of the following:

Item No.	Quantity	Description	Requirement
1	1 each	Trailer, Van V-83( )/M	3.5
2	1 each	Air Conditioner, Type MA-5	3.5
3	1 each	Terminal Box J-740/M	3.5
4	1 each	Inside signal wire terminal	3.5
5	1 each	Distribution Box J-924/MRN-12	3.5
6	1 each	Rack, Electrical Equipment MT-1885/MRN-13	3.5
7	1 each	Control, Transmitter C-2385/URT	3.5
8	1 each	Clock	3.6
9	1 each	Flashlight TL-122	3.6
10	1 each	Light	3.6
11	1 each	Cord, CD-307-A	3.6
12	12 each	Ground Rod MX-148/G	3.6
13	1 each	Headset HS-33-A	3.6
14	1 each	Reel RL-39-B	3.6
15	1 each	Tape	3.6
16	1 each	Frequency Meter BC-221	3.6
17	1 each	Rectifier Power Unit RA-133-A	3.6
18	1 each	Test Set, Electron Tube TV-7/U	3.6
19	1 each	Radio Frequency Indicator TS-446/U	3.6
20	1 each	Multimeter AN/PSM-6	3.6
21	1 each	Tool Equipment TE-113	3.6
22	1 each	Fire extinguisher	3.6
23	1 each	Chair	3.6
24	1 each	Cabinet, Power Supply CY-1198/URN-5	3.6
25	1 each	Transmitter, Radio T-357/URN-5	3.6
26	2 each	Mast AB-86/GRA-4	3.6
27	1 each	Modulator-Power Supply MD-174/URN-5	3.6

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Item No.	Quantity	Description	Requirement
28	1 each	Coupler, Antenna CU-317/URN-5	3.6
29	1 each	Control, Transmitter C-1107/URN-5	3.6
30	1 each	Microphone, Dynamic M-37/U	3.6
31	1 each	Cord CG-275/U (10 feet 0 inch)	3.6
32	1 each	Telephone Set TA-312/PT	3.6
33	1 each	Battery BA-1030/U	3.6
34	8 each	Wire, electrical	3.6
35	3 each	Reel RL-122/GRA-4	3.6

3.3 GENERAL SPECIFICATION.- The requirements of MIL-E-4158 apply as requirements of this specification. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern. Exceptions and additions to the general specification shall be as follows:

a. Ambient temperature

(1) Operating: Minus 40°C to plus 65°C

(2) Nonoperating: Minus 62°C to plus 71°C

b. Winds

Operating and Nonoperating: Up to 56 knots without requiring guys to stabilize the trailer. Up to 109 knots without guys

c. Barometric pressure

(1) Operating: From 30 to 24 inches of mercury (sea level to approximately 6 000 feet above).

(2) Nonoperating: From 30 to 5.54 inches of mercury (sea level to approximately 40,000 feet above sea level).

d. Ice load

Operating and Nonoperating: 2 inches of ice measured radial to exposed surfaces

3.3.1 SERVICE CONDITIONS (MECHANICAL).- The radio beacon shall be constructed so that no fixed part will become loose, no movable part or permanently set adjustment be shifted in setting or position, no component will be damaged, and no degradation will be caused in the performance of its components when subjected to high impact shock, bumping, dropping, vibration and other rigorous service conditions encountered when transported in all kinds of weather by vehicles over rough terrain, by air transport or by train or ship.

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3.3.2 SERVICE CONDITIONS (ELECTRICAL).- The equipment shall be designed to operate from power sources with the following characteristics:

a.	Potential:	120/240 volts plus or minus 5 percent
b.	Frequency:	50 to 60 cps
c.	Phase:	Single phase
d.	Harmonic content:	Not to exceed 5 percent

3.3.3 SERVICE LIFE.- The equipment shall be capable of operating 23 hours per day for a period of 2000 hours with only normal maintenance and without major overhaul. The operational period may start after an initial storage period not exceeding 2 years.

3.3.4 INTERCHANGEABILITY.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.4 DESIGN AND CONSTRUCTION.- The radio beacon shall be constructed in accordance with the design contained on Craig System Drawings as listed on 72R70. Where the requirements of this specification and the drawings conflict, the requirements of the drawings shall govern.

3.4.1 INTERCONNECTING CABLES, CORDS, WIRING AND CONNECTORS.- All cables, cords, wiring, connectors, terminal panels, and junction boxes not furnished by the Government and necessary to interconnect the components of the radio beacon, whether they are mounted inside or outside the trailer van in their operating position, shall be provided as part of radio beacon. The contractor shall utilize Government-furnished cables to the greatest extent possible. All electrical wiring shall be protected by suitable metallic conduits or ducts.

3.5 DETAILS OF COMPONENTS.- The details of components of the radio beacon shall be in accordance with Craig System Drawings as listed on 72R70.

3.6 GOVERNMENT-FURNISHED PROPERTY.- When the contract or purchase order so provides, the Governement will furnish the following to the contractor for use in making the radio beacon:

Item No.	Stock No.	Part No.	Description	Quantity
1	6244-22809-A	AN5743-1A	Clock, 8-day	1 each
2	7700-331862		Flashlight TL-122	1 each
3	7700-529880		Light, Extension, 50 foot	1 each
4	6135-120-1020 (Sig FSC)		Battery BA-1030/U	2 each
5	1790-321018000		Cord CD-307-A 5 foot, 5 inches	1 each
6			Ground Rod MX-148/G	12 each
7	1790-207610005		Headset HS-33-A	1 each

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Item No.	Stock No.	Part No.	Description	Quantity
8	5805-693-7113	605070	Telephone TA-312/PT	1 each
9	3895-498-8343 (Sig FSC)	6H3039	Reel RL-39-E	1 each
10	7900-774180		Tape, measuring, 100 foot	1 each
11	7CAC-318208-45		Frequency Meter BC-221	1 each
12	7CAC-652885		Rectifier Power Unit RA-133-A	1 each
13	7CAC-801318-5725		Test Set, Electron Tube TV-7/U, MIL-T-12424	1 each
14	7CAC-439725		Radio Frequency Indicator TS-445/U	1 each
15	7CAC-589073-3		Multimeter AN/PSM-6, MIL-M-4552	1 each
16	1800-679406000	6R38113	Tool Equipment TE-113	1 each
17	8200-245016		Fire Extinguisher, CO_2, 15 pounds	1 each
18	3500-194810		Chair, metal, adjustable	1 each
19	1700-018519536		Cabinet-Power Supply CY-1198/URN-5, MIL-R-15710	1 each
20	1700-019634007		Transmitter, Radio T-357/URN-5, MIL-R-15710	1 each
21	5985-257-2927	2A282-86	Mast AB-86/GRA-4	2 each
22	5985-497-8765 (Sig FSC)	2A3155-122	Reel RL-122/GRA-4	3 each
23			Modulator-Power Supply MD-174/URN-5 MIL-R-15710	1 each
24	1700-016095450		Coupler, Antenna CU-317/URN-5	1 each
25	1700-016095280		Control, Transmitter C-1107/URN-5	1 each
26	5965-280-7807 (AF FSC)		Microphone, Dynamic M-37/U MIL-R-15710	1 each
27			Cord CG-275/U (10 feet, 0 inch), Drawing No. SC-D-15046	1 each
28			Wire, Electrical, No. 14, AWG, Bare Copper (100 feet, 0 inch)	8 each

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS.- The inspection and testing of the radio beacon shall be classified as follows:

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a. Acceptance tests. . . . . See 4.2
b. Preproduction testing . . . . . See 4.5

4.2 ACCEPTANCE TESTS.- Acceptance tests shall consist of:

a. Individual tests
b. Periodic sampling plans and tests

4.2.1 INDIVIDUAL TESTS.- Each radio beacon shall be subjected to the following tests described under 4.4 "TEST METHODS" of this specification:

a. Examination of Product
b. Electrical inspection test

In addition, each radio beacon shall be subjected to any other tests specified in 4.4 which the procuring activity considers necessary to determine compliance with the requirements of this specification.

4.2.2 PERIODIC SAMPLING PLANS AND TESTS

4.2.2.1 PERIODIC SAMPLING PLAN.- One radio beacon shall be selected at random from each 50 radio beacon manufactured during the production period and subjected to the following tests:

Performance test . . . . . See 4.4.7

4.2.2.2 REJECTION AND RETEST.- When one or more items from a lot fail to meet the specification, acceptance of all items in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.2.2.3 INDIVIDUAL TESTS MAY CONTINUE.- For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.2.2.4 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.3 TEST CONDITIONS.- Unless otherwise specified, all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

4.4 TEST METHODS

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4.4.1 EXAMINATION OF PRODUCT.- Each equipment shall be given a thorough mechanical and visual inspection and test to determine that the quality of material and workmanship is in compliance with the requirements of this specification. Particular attention shall be given to the following:

a. Completeness
b. Nameplates, identification markings and labels
c. Ease of operation of gears, adjustable and sliding parts, thumb screws, controls and switches
d. Finishes
e. Welded joints
f. Check of solder joints
g. The fit of components in their respective positions
h. Check of mounting means
i. Check of lubrication and rust prevention
j. Check of safety features and interlocks
k. Loose fastening and securing devices or parts
l. Accessibility of components and parts for servicing
m. Cable runs between components including plugs and receptacles
n. Grounding connections
o. Over-all dimensions check
p. Weight check
q. Check of provisions for securing components during transit
r. Check of shock-mounting provisions for components
s. Check of trailer van illumination
t. Check of heating and ventilating provisions
u. Availability and accessibility of space for operator
v. Facilities for storage of tools, test equipment, instructional literature and spare parts
w. Other visual defects

4.4.2 ELECTRICAL INSPECTION.- Each equipment shall be given thorough electrical inspection tests as are considered necessary to determine that all circuits are inherently sound so that over-all performance of the equipment in compliance with the requirements of this specification shall be obtained. The electrical tests shall include but not necessarily be restricted to the following.

4.4.2.1 CONTINUITY.- Each electrical component and each cord and cable shall be given a continuity test to ascertain if it is wired and connected correctly and that good electrical contact is obtained.

4.4.2.2 OPERATING VOLTAGES.- The operating voltage at all important points, shall be checked for conformance with those shown on the circuit labels and schematic drawing. This shall be done with all controls set for normal operation of the equipment.

4.4.2.3 With 120/240 volt, single phase power connected to the power input receptacle and with all circuit breakers closed, all electronic equipment shall be turned on and operated in its normal operating position.

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The operation of all electronic equipment shall be checked. Locally generated test signals may be used to check the operation of the equipment. Signals shall be inserted in the outside signal terminal panel to check circuit continuity through to the output terminals. This test shall demonstrate that all the equipment is wired and connected correctly for proper operation.

4.4.3 ENVIRONMENTAL TESTS.- The equipment shall be subjected to the following environmental conditions in accordance with MIL-E-5272 with exceptions as specified:

a. High temperature test (Procedure I).- Maintain the equipment at a temperature of plus 71°C for 4 hours, reduce the temperature to plus 65°C for 4 hours, then operate the equipment and compare results

b. Low temperature test (Procedure II).- Operating temperature shall be minus 40°C. Nonoperating temperature shall be minus 62°C

c. Altitude test (operating) (Procedure I).- Perform at an absolute pressure of 24 inches of mercury (pressure at approximately 6000 feet above sea level) on components selected by the procuring activity

d. Altitude test (nonoperating).- Repeat Procedure I on components selected by the procuring activity at an absolute pressure of 5.54 inches of mercury (pressure at approximately 40,000 feet above sea level) for 4 hours, but do not operate the equipment.

4.4.3.1 After completion of each test above, the performance of the equipment shall not be below the minimum requirements specified herein.

4.4.4 BOUNCE TEST.- The radio beacon assembled in its transportable condition shall be capable of withstanding the bounce test in accordance with MIL-T-4807, method 4C without breakage of any part, permanent deformation of any part, loosening of threaded or other parts, other mechanical damage, or electrical damage. In addition, there shall be no evidence of collision between any parts of the radio beacon. Upon completion of the test, the radio beacon shall be subjected to an operational test to determine that there has been no degradation in its performance.

4.4.5 ASSEMBLY AND DISASSEMBLY TEST.- The radio beacon, after being assembled for operation, shall be disassembled and placed into its road transportable condition. After the disassembly has been completed, the radio beacon shall be reassembled into its operating condition and operated satisfactorily for 1 hour. A record shall be kept of the time required for disassembling the system for road transport and for reassembling the radio beacon for operation.

4.4.6 AIRCRAFT LOADING TEST.- The radio beacon shall be disassembled into its air transportable condition. The complete radio beacon in its air transportable condition shall then be loaded into a Type C-119 aircraft. A record shall be kept of the time required to accomplish the

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loading in the aircraft. This test shall demonstrate that the radio beacon has the necessary form factor for loading into the aircraft. The Type C-119 aircraft will be furnished by the Government.

4.4.7 PERFORMANCE TESTS.- Each equipment shall be given thorough performance tests as are considered necessary to determine that the equipment meets the performance requirements specified herein.

4.5 PREPRODUCTION TESTING

4.5.1 PREPRODUCTION TEST SAMPLES.- The preproduction test sample shall consist of models representative of production equipment. One sample shall be tested at a laboratory designated by the procuring activity, or when stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.5.2 PREPRODUCTION TEST REPORTS.- When the contractor performs the preproduction test, he shall prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.5.3 PREPRODUCTION TESTS.- The preproduction tests shall consist of all the tests described under 4.4 "TEST METHODS."

5. PREPARATION FOR DELIVERY

5.1 PREPARATION FOR SHIPMENT.- The radio beacon shall be prepared and marked for shipment in accordance with MIL-P-7936. The shipment marking nomenclature shall be as follows: "Beacon, Radio AN/MRN-13( )."

6. NOTES

6.1 INTENDED USE.- Beacon, Radio AN/MRN-13( ) is intended for use as a homing facility in guiding aircraft to a desired location.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Level of preservation, packaging and packing required. (See 5.1.)

c. Whether the preproduction test will be conducted by the contractor or the procuring activity. (See 4.5.1.)

d. Conditions under which Government-furnished property will be made available to the contractor. (See 3.6.)

e. Conditions under which design approval will be granted.

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f. A complete set of revised drawings shall not be required. Only drawings revised as a result of design changes made during the contract should be furnished to the procuring activity by the contractor.

6.3 NOMENCLATURE.- The parentheses appearing in the nomenclature will be dropped or replaced by a letter, to indicate the particular design. Nomenclature for the individual components, where required, will be furnished upon application to the procuring activity.

6.4 DRAWINGS.- Upon specific request of the contractor to the procuring activity, a copy of the applicable drawings of the equipment, if available, may be furnished for informational purposes only.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related therto.

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